

B.Sc. (Hons.) Semester III Examination, 2013-14
STATISTICS

Paper No.: STB-301: Statistical Inference

Time: Three hours

Full Marks: 70

Note: (i) Attempt any five questions. (ii) All questions carry equal marks.

1. (a) Define a parameter and a statistic and give some examples of both. What do you mean by a random sample of fixed size from a probability distribution?
- (b) Let X be a random variable with expectation $E(X) = \mu$ and variance $V(X) = \sigma^2$. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from the distribution of X . Then show that:

$$(i) E(\bar{X}) = \mu, \quad (ii) V(\bar{X}) = \sigma^2/n$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

2. (a) What is meant by the sampling distribution of a statistic and its standard error? Let $X_1, X_2, \dots, X_n$ be a random sample of size n where the random variables X_i ($i = 1, 2, \dots, n$) are independent and follow a Poisson distribution with parameter λ_i .
 - i. What is the distribution of the random variable $S_n = X_1 + X_2 + \dots + X_n$?
 - ii. If $\bar{X} = S_n/n$, find $E(\bar{X})$ and the standard error of $\bar{X}$.
- (b) Let the independent random variables $X_1, X_2, \dots, X_n$ be distributed as $N(\mu, \sigma^2)$. Let $X = \sum_{i=1}^n \alpha_i X_i$ and $Y = \sum_{j=1}^n \beta_j X_j$, where α_i and β_j are constants. Determine the probability distribution of X , Y , and $X + Y$.
3. (a) In reference to testing of hypothesis, define the following terms with suitable examples:
 - i. Null and alternative hypotheses
 - ii. Type-I and Type-II errors
 - iii. Critical and acceptance regions
- (b) Define (i) simple and composite hypotheses, (ii) level of significance, and (iii) power of the test. Classify the following four sets of hypotheses for testing a mean as simple or composite:
 - i. $H_0 : \mu = \mu_0, \quad H_1 : \mu = \mu_1$
 - ii. $H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0$
 - iii. $H_0 : \mu = \mu_0, \quad H_1 : \mu > \mu_1$
 - iv. $H_0 : \mu < \mu_0, \quad H_1 : \mu > \mu_1$
4. (a) Let X_i ($i = 1, 2, \dots, n$) be independent random variables such that X_i follows a normal distribution with mean μ_i and variance σ_i^2 . Find the distribution of the statistic $\sum_{i=1}^n \left(\frac{X_i - \mu_i}{\sigma_i} \right)^2$. Name the distribution, and find its mean and variance.
- (b) State and prove the additive property of chi-square variates.
5. (a) Define Student's t -statistic and give its probability distribution. Obtain the expression of the even-order central moments of the distribution and hence find the expressions for μ_2 and μ_4 .
- (b) What is F -statistic? Write its probability distribution and hence find its mean.

6. (a) A bird watcher sitting in a park has spotted birds belonging to 6 categories with the following frequencies:

Category	1	2	3	4	5	6
Frequency	6	7	13	17	6	5

Test at a 5% level of significance whether the data is compatible with the assumption that the park is visited by birds in the proportion 1 : 1 : 2 : 3 : 1 : 1. Given $\chi^2_{0.05,6} = 12.592$, $\chi^2_{0.05,5} = 11.07$, $\chi^2_{0.05,4} = 9.488$.

- (b) Samples of two types of electric light bulbs were tested for length of life and the following data were obtained:

	Type I	Type II
Sample Size	8	7
Sample Mean	1234 hours	1036 hours
Sample S.D.	36 hours	40 hours

Is the difference in means sufficient to warrant that Type I is superior to Type II regarding length of life? Given $t_{0.05,15} = 1.75$, $t_{0.05,14} = 1.76$, $t_{0.05,13} = 1.77$.

7. (a) What do you understand by non-parametric tests? Describe the run test for testing randomness. Apply the Mann-Whitney-Wilcoxon U test to the following data to test if both populations have the same distribution:

Experimental scores	110	70	53	51	
Controlled scores	78	64	75	45	82

- (b) What are order statistics? Explain with an example. Obtain the distribution function and density function of the k^{th} order statistic. Let $X_1, X_2, \dots, X_n$ be n i.i.d. variates with distribution function $F(x) = 1 - e^{-\lambda x}$ ($\lambda > 0, x > 0$). Find the density function of the smallest order statistic.

8. Write short notes on any **three** of the following:

- (a) Fisher's Z-transformation and its uses
- (b) Weak law of large numbers
- (c) χ^2 -test for independence of attributes
- (d) Sign test
- (e) Central limit theorem and its applications in testing of significance

External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>